

As questões de 1 a 7 referem-se ao texto a seguir:

FRAYING AT THE EDGES: A LIFE-CHANGING DIAGNOSIS

IT BEGAN WITH what she saw in the bathroom mirror. On a dull morning, Geri Taylor padded into the shiny bathroom of her Manhattan apartment. She casually checked her reflection in the mirror, doing her daily inventory. Immediately, she stiffened with fright.

Huh? What?

She didn't recognize herself.

She gazed saucer-eyed at her image, thinking: Oh, is this what I look like? No, that's not me. Who's that in my mirror?

This was in late 2012. She was 69, in her early months getting familiar with retirement. For some time she had experienced the sensation of clouds coming over her, mantling thought. There had been a few hiccups at her job. She had been a nurse who climbed the rungs to health care executive. Once, she was leading a staff meeting when she had no idea what she was talking about, her mind like a stalled engine that wouldn't turn over.

"Fortunately I was the boss and I just said, 'Enough of that; Sally, tell me what you're up to,'" she would say of the episode.

Certain mundane tasks stumped her. She told her husband, Jim Taylor, that the blind in the bedroom was broken. He showed her she was pulling the wrong cord. Kept happening. Finally, nothing else working, he scribbled on the adjacent wall which cord was which.

Then there was the day she got off the subway at 14th Street and Seventh Avenue unable to figure out why she was there.

So, yes, she had had inklings that something was going wrong with her mind. She held tight to these thoughts. She even hid her suspicions from Mr. Taylor, who chalked up her thinning memory to the infirmities of age.

"I thought she was getting like me," he said. "I had been forgetful for 10 years."

But to not recognize her own face! To Ms. Taylor, this was the "drop-dead moment" when she had to accept a terrible truth. She wasn't just seeing the twitches of aging but the early fumes of the disease.

She had no further issues with mirrors, but there was no ignoring that something important had happened. She confided her fears to her husband and made an appointment with a neurologist.

"Before then I thought I could fake it," she would explain. "This convinced me I had to come clean."

In November 2012, she saw the neurologist who was treating her migraines. He listened to her symptoms, took blood, gave her the Mini Mental State Examination, a standard cognitive test made up of a set of unremarkable questions and commands. (For instance, she was asked to count backward from 100 in intervals of seven; she had to say the phrase: "No ifs, ands or buts"; she was told to pick up a piece of paper, fold it in half and place it on the floor beside her.)

He told her three common words, said he was going to ask her them in a little bit. He emphasized this by pointing a finger at his head — remember those words. That simple. Yet when he called for them, she knew only one: Beach. In her mind, she would go on to associate it with the doctor, thinking of him as Dr. Beach.

He gave a diagnosis of mild cognitive impairment, a common precursor to Alzheimer's disease. The first label put on what she had. Even then, she understood it was the footfall of what would come. Alzheimer's had struck her father, a paternal aunt and a cousin. She long suspected it would eventually find her.

Fonte: <http://www.nytimes.com/interactive/2016/05/01/nyregion/living-with-alzheimers.html?action=click&contentCollection=Americas&module=Trending&version=Full®ion=Marginalia&pgtype=article>. (acesso em 1/05/2016).

Questão 1. Quanto ao gênero textual, o texto pode ser classificado como

- A () sinopse de romance. B () romance autobiográfico. C () romance biográfico.
D () laudo psiquiátrico. E () resenha médica.

Questão 2. Quanto à narrativa, o texto é apresentado

- A () por narrador externo, que utiliza discursos direto e indireto.
B () pela personagem principal, que relata ao leitor suas limitações cognitivas.
C () por uma personagem que traz também a voz da personagem principal.
D () pelo médico que busca apresentar informações sobre Geri Taylor.
E () pela própria narradora, quando ainda tinha consciência dos fatos de sua vida.

Questão 3. De acordo com o texto,

- A () Geri Taylor ficou abatida com o diagnóstico de Alzheimer e declarou ter sido nocauteada pela doença.
- B () Jim Taylor procurou o médico para conversar sobre os esquecimentos da esposa.
- C () o neurologista teve certeza da doença de Geri Taylor porque ela não conseguiu contar de 0 a 100.
- D () Geri Taylor foi diagnosticada com um problema cognitivo que normalmente precede o Alzheimer.
- E () Jim Taylor preferiu não contar ao médico sobre as doenças degenerativas na família da esposa.

Questão 4. Marque a opção correta quanto aos procedimentos solicitados pelo neurologista a Geri Taylor.

- A () Contar de 0 a 100 em intervalos de 7 em 7.
- B () Repetir cinco palavras em intervalos curtos de tempo.
- C () Dobrar uma folha de papel ao meio e posicioná-la no chão.
- D () Elaborar perguntas e respondê-las em sequência.
- E () Citar frases populares bem conhecidas.

Questão 5. Marque a opção que contém a principal causa da decisão de Geri Taylor em buscar diagnóstico médico.

- A () Dificuldade de reconhecer sua própria imagem diante do espelho.
- B () Esquecimento de palavras importantes em reuniões de trabalho.
- C () Dificuldades recorrentes para lidar com a persiana do quarto do casal.
- D () Sensação de estar perdida em pontos da cidade por onde tinha costume de circular.
- E () Histórico de Alzheimer em pessoas da família, como o próprio pai e parentes paternos.

Questão 6. Marque a opção em que o item sublinhado **NÃO** é classificado como um advérbio.

- A () She casually checked her reflection in the mirror, [...] (linha 2)
- B () “Fortunately I was the boss and I just said, [...] (linha 13)
- C () Finally, nothing else working, he scribbled on the adjacent wall which cord was which. (linhas 16-17)
- D () She wasn’t just seeing the twitches of aging but the early fumes of the disease. (linha 25)
- E () She long suspected it would eventually find her. (linha 40)

Questão 7. Marque a opção em que o item sublinhado é um qualificador.

- A () Once, she was leading a staff meeting when she had no idea [...] (linhas 10-11)
- B () So, yes, she had had inklings that something was going wrong [...] (linha 20)
- C () [...] Mr. Taylor, who chalked up her thinning memory to the infirmities of age. (linhas 21-22)
- D () [...] but there was no ignoring that something important had happened. (linhas 26-27)
- E () He emphasized this by pointing a finger at his head – remember those words. (linhas 34-35)

As questões de 8 a 12 referem-se ao texto a seguir:

**INSIDE THE BILL GATES-BACKED ACCELERATOR THAT'S TRAINING
THE NEXT GENERATION OF VENTURE CAPITALISTS**

Lauren Gensler, FORBES STAFF

- 1 In an airy converted furniture store in Seattle's Pioneer Square neighborhood, five novice impact
- 2 fund managers from Zimbabwe, Guatemala and the Netherlands are rehearsing the sales pitches they'll
- 3 make the next day to 60 mostly institutional investors, representing \$10 billion in capital.
- 4 The presentations will be a graduation ceremony of sorts. Despite their impressive resumes, the five
- 5 men have just completed a four-week boot camp covering everything from term sheets, accounting and
- 6 mezzanine debt structures to dealing with corruption to defining and marketing their brands. They'll head
- 7 home with golden contacts (investor cocktail hours were built into the packed schedule) and a commitment for
- 8 up to \$500,000 in seed capital from Capria Accelerator, a first-of-its-kind venture whose initial investors
- 9 include Microsoft cofounder (and world's richest man) Bill Gates.

10 One of those rehearsing is Patrick Makanza, 51, an M.B.A. and veteran of Unilever and Barclays
11 Bank, who quit a cushy job at a top Zimbabwe private equity firm and launched Vakayi Capital. The first fund
12 being formed by Vakayi (which means “to build”) will back for-profit businesses providing essential services in
13 Zimbabwe, which has per capita gross domestic product of about \$1,000. That fund will make loans (with an
14 option to convert some to equity) for an average of four years to small and medium-size businesses that want
15 to expand and can’t get adequate bank financing. Among potential investments: an eye clinic that’s building a
16 new operating room so it can double its daily procedures and bring down the cost of cataract surgery; an
17 education microlender; and a builder of low-cost housing.

18 The next day, in their presentation, Makanza and his Vakayi cofounder tackle head-on the tough
19 issues they might be asked about, including Zimbabwe’s endemic corruption, economic challenges and
20 currency dramas—it squelched hyperinflation in 2009 by switching to foreign currencies, primarily the U.S.
21 dollar. But in the question period Tracy Washington, principal investment officer for the International Finance
22 Corp.’s global private equity funds, lobs a personal query at Makanza, a father of four who is partial to
23 conservative business suits and golf. “With your resume”, she asks, “why get involved with so risky an
24 enterprise, and will you stick to it?” Makanza responds that he worked in venture capital back in the 1990s
25 and came to miss the highs and lows of investing in early-stage entrepreneurs. “I still have at least ten years
26 to do this. ... It’s a real roller-coaster lifestyle. But I enjoyed it, and I want to have more of that experience
27 again.”

28 Impact investing – which aims to produce both financial and social or environmental returns – is in
29 vogue. Big names in finance, from BlackRock to Goldman Sachs to Bank of America Merrill Lynch, have been
30 piling in recently, seeing it as a way to appeal to the socially conscious Millennials now building and inheriting
31 wealth.

32 But this alternative asset class is still small – \$77 billion invested worldwide, according to a new
33 survey from the Global Impact Investing Network. To grow, it needs experienced, hands-on fund managers,
34 and those are in short supply, particularly in areas with the greatest needs, such as sub-Saharan Africa.

Fonte: <http://www.forbes.com/sites/laurengensler/2016/06/15/capria-bill-gates-impact-investing-accelerator/#56afe1ab1dc4>.
(Acesso em 07/07/2016).

Questão 8. De acordo com o texto,

- A () cinco profissionais da área financeira, de diferentes partes do mundo, preparam-se para apresentar suas propostas a investidores.
- B () o curso sobre gestão financeira realizado em Seattle, de caráter eliminatório, teve duração de quatro semanas.
- C () contatos profissionais são mais importantes do que o domínio de conhecimentos específicos e teóricos.
- D () a comparação entre fundos de investimentos e a expressão "montanha russa" deve-se ao panorama econômico dos últimos dez anos.
- E () investimento de impacto é uma tendência que agrega obrigatoriamente retornos financeiro, social e ambiental.

Questão 9. É **INCORRETO** afirmar que Patrick Makanza

- A () é um profissional experiente que deixou uma carreira bem sucedida para se arriscar em um novo empreendimento.
- B () faz questão de envolver os quatro filhos na gestão de todos os seus empreendimentos, apesar de ser um gestor conservador.
- C () lançou Vakayi Capital com o objetivo de oferecer empréstimos a empresas que querem expandir, mas não conseguem financiamento bancário adequado.
- D () participa de um evento na cidade de Seattle, preparando-se para responder perguntas sobre corrupção e riscos financeiros no Zimbábue.
- E () busca investir em empresas que prestem serviços essenciais no Zimbábue, como, por exemplo, na área de saúde, educação e habitação.

Questão 10. Marque a opção que substitui o trecho sublinhado, mantendo o mesmo sentido.

“Despite their impressive resumes, the five men have just completed a four-week boot camp covering everything from term sheets, [...]”

- A () Once they have impressive resumes,
- B () Besides they have impressive resumes,
- C () Since their impressive resumes,
- D () Because of their impressive resumes,
- E () Although they have impressive resumes,

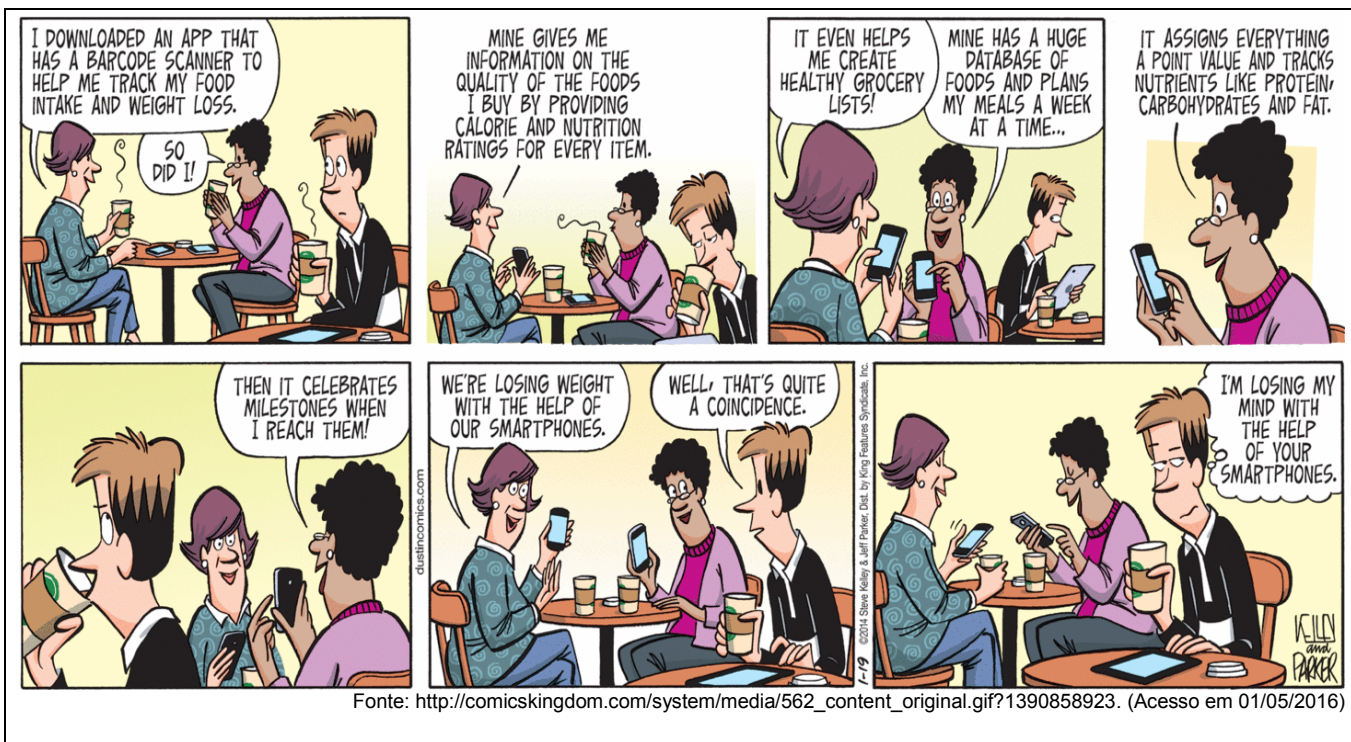
Questão 11. Marque a opção em que o item sublinhado **NÃO** exerce a função de agente da oração.

- A () [...] Patrick Makanza, 51, an M.B.A. and veteran of Unilever and Barclays Bank, who quit a cushy job at a top Zimbabwe private equity firm [...] (linhas 10-11)
- B () That fund will make loans [...] for an average of four years to small and medium-size businesses that want to expand [...] (linhas 13-14-15)
- C () [...] Tracy Washington [...] lobs a personal query at Makanza, a father of four who is partial to conservative business suits and golf. (linhas 21-22-23)
- D () Makanza responds that he worked in venture capital back in the 1990s and came to miss the highs and lows of investing in early-stage entrepreneurs. (linhas 24-25)
- E () Impact investing – which aims to produce both financial and social or environmental returns – is in vogue. (linhas 28-29)

Questão 12. O termo **whose** em: “They’ll head home with golden contacts (investor cocktail hours were built into the packed schedule) and a commitment for up to \$500,000 in seed capital from Capria Accelerator, a first-of-its-kind venture whose initial investors include Microsoft cofounder (and world’s richest man) Bill Gates.”, refere-se a

- A () golden contacts.
- B () a commitment.
- C () seed capital.
- D () Capria Accelerator.
- E () Bill Gates.

A tirinha a seguir mostra um diálogo entre duas pessoas, com a participação de um terceiro interlocutor. Analise-a e responda as questões de 13 a 16.



Questão 13. A terceira pessoa mostra-se incomodada devido à/ao

- A () incoerência das informações tratadas na conversa.
- B () excesso de informações e à falta de privacidade.
- C () impossibilidade de adquirir um *smartphone*.
- D () fato de ser ignorada pelas pessoas que estão conversando.
- E () resistência pessoal quanto ao uso de aplicativos para emagrecer.

Questão 14. Depreende-se da leitura que os aplicativos

- A () podem ser instalados em qualquer modelo de aparelho celular.
- B () oferecem exatamente as mesmas facilidades às duas usuárias.
- C () registram informações nutricionais nos rótulos dos alimentos.
- D () impedem o consumo de alimentos de má qualidade nutricional.
- E () auxiliam as usuárias no controle da perda de peso.

Questão 15. As palavras utilizadas como referentes aos **aplicativos** são

- A () that, mine, it.
- B () it, even, mine.
- C () that's, it, them.
- D () that, me, mine.
- E () them, that, it.

Questão 16. Marque a opção em que os itens da tirinha possuem a mesma classificação gramatical.

- A () then - quite
- B () intake - milestones
- C () loss - huge
- D () like - reach
- E () providing - ratings

As questões de 17 a 20 referem-se ao texto a seguir:

**STARSHOT PROJECT: STEPHEN HAWKING AND MARK ZUCKERBERG LAUNCH
MOST AMBITIOUS ALIEN-FINDING PROJECT EVER**

- 1 Tiny rockets are going to be sent into space to study the far universe in the most ambitious space exploration
- 2 project in history.
- 3 Scientists including Stephen Hawking and backers such as internet investor Yuri Milner and Mark Zuckerberg
- 4 will send "nano craft" deep into space to explore the most remote regions that humans have ever seen, by far.
- 5 The hugely ambitious project could reveal deep secrets of the universe and will allow people to photograph
- 6 one of the most likely places to hold life on other worlds.
- 7 Professor Hawking said at the event: "What makes us unique is transcending our limits. Gravity pins us to the
- 8 ground, but I just flew to America.
- 9 "How do we transcend these limits? With our minds and our machines.
- 10 "The limit that confronts us now is the great void between us and the stars. But now we can transcend it, with
- 11 light beams, light sails, and the lightest spacecraft ever built. Today we commit to this next great leap into the
- 12 cosmos, because we are human and our nature is to fly."
- 13 The Starshot Project hopes to get the tiny robots out to the Alpha Centauri star system, 25 trillion miles away.
- 14 Getting there through normal means would take 30,000 years – but the new project hopes that using the tiny
- 15 rockets will allow them to get there in just 20.
- 16 Scientists think that the Alpha Centauri system might well have an Earth-like planet that could be found in its
- 17 "habitable zones". The craft will be able to take pictures of those – a potential way that they might find life on
- 18 other worlds.
- 19 The crafts will be "gram-scale nano craft", according to Yuri Milner, which will make their way through space
- 20 using a "sail pushed by a light beam". Their design will allow them to fly at 25 per cent of light speed.
- 21 Those craft will be able to send back images of possible planets and other scientific data, according to the
- 22 scientists behind it.
- 23 "The human story is one of great leaps," Dr Milner said. "Today we are preparing for the next great leap – to
- 24 the stars.
- 25 "Can we literally reach the stars, and can we do it in our lifetime?"
- 26 The tiny rockets are made up of computers that can be mounted to a tiny "wafer". Shrinking computer
- 27 components mean that all of the necessary parts – cameras, thrusters, power supply and navigation
- 28 equipment – can all be mounted on a tiny plate that will be a fully functional space probe.

29 Before those are built, the project will have to create all of the important parts on the ground. That includes
30 the construction of a light-beamer that can power the rockets and a "mothership" that will be able to carry
31 them all out into space and launch them.
32 Because of economies of scale and the decreasing price of computer components, the team will eventually
33 be able to send out the rockets for just a few hundred thousand dollars, they said.

Fonte: <http://www.independent.co.uk/news/science/starshot-project-stephen-hawking-and-mark-zuckerberg-to-send-tiny-rockets-to-alpha-centauri-in-most-a6981101.html> (acesso em 03/05/2016).

Questão 17. De acordo com o texto, o principal objetivo do Projeto *Starshot* é

- A () enviar robôs minúsculos para encontrar e investigar vida alienígena inteligente em galáxias distantes.
- B () encontrar planetas semelhantes à Terra, para os quais humanos devam ser transportados dentro de vinte anos.
- C () enviar "nanofoguetes" para a galáxia mais distante do sistema solar, aonde nenhum ser humano seria capaz de chegar.
- D () enviar foguetes minúsculos ao sistema estelar Alfa Centauri a fim de obter imagens de seus planetas e outros dados científicos.
- E () alterar as leis da gravidade e transcender os limites da ciência atual com nossas mentes e máquinas de forma nunca antes imaginada.

Questão 18. De acordo com o texto, é correto afirmar que

- A () os "nanofoguetes" percorrerão o trajeto em um centésimo do tempo que uma espaçonave comum faria.
- B () é esperado que os "nanofoguetes" cheguem ao sistema estelar Alfa Centauri em apenas duas décadas.
- C () o sistema Alfa Centauri se encontra a 25 trilhões de quilômetros de distância da Terra.
- D () o Projeto *Starshot* levará robôs minúsculos ao sistema estelar Alfa Centauri em 30.000 anos.
- E () a "nave-mãe" navegará a 25% da velocidade da luz até o sistema Alfa Centauri.

Questão 19. Quanto à execução do Projeto *Starshot*, o texto afirma que

- A () os "nanofoguetes" serão constituídos por circuitos e baterias autocarregáveis, a fim de facilitar o arquivamento dos dados.
- B () a última etapa para a criação dos "nanofoguetes" será definir o tamanho da nave-mãe e de todos os seus componentes.
- C () as câmeras, os propulsores, as fontes de energia e os equipamentos de navegação foram desenvolvidos em um projeto anterior.
- D () os investidores em componentes para computadores comercializarão os "nanofoguetes" a fim de lucrar centenas de milhares de dólares.
- E () a economia de escala é um dos fatores que poderá permitir o envio dos "nanofoguetes" ao espaço por algumas centenas de milhares de dólares.

Questão 20. Na frase "But now we can transcend it, with light beams, light sails, and the lightest spacecraft ever built." (linhas 10 e 11), é correto afirmar que

- A () o adjetivo *light* é usado duas vezes no grau normal e uma no comparativo.
- B () a palavra *light* significa *luz* e tem função de agente dos verbos *beam* e *sail*.
- C () *the lightest* significa *a mais leve* e está flexionada no grau superlativo.
- D () somente nas duas primeiras ocorrências, o significado de *light* é *leve*.
- E () as três ocorrências da palavra *light* têm o mesmo significado.

FIM DA PROVA

**PARE AQUI SE AINDA NÃO QUISE
VER AS RESPOSTAS.**

**O GABARITO COMEÇA NA PRÓXIMA
PÁGINA.**



INSTITUTO TECNOLÓGICO DE AERONÁUTICA

VESTIBULAR 2017

GABARITO

Física		Inglês		Português		Matemática		Química	
1	A	1	C	21	E	1	A	1	A
2	A	2	A	22	E	2	A	2	B
3	D	3	D	23	D	3	D	3	D
4	E	4	C	24	A	4	B	4	D
5	B	5	A	25	C	5	C	5	A
6	E	6	D	26	A	6	C	6	D
7	(*)	7	C	27	C	7	C	7	C
8	D	8	A	28	E	8	E	8	A
9	D	9	B	29	C	9	A	9 (**)	E
10	B	10	E	30	E	10	C	10	E
11	C	11	D	31	A	11	A	11	E
12	C	12	D	32	C	12	E	12	C
13	C	13	B	33	B	13	A	13	E
14	B	14	E	34	A	14	B	14	D
15	A	15	A	35	E	15	D	15	C
16	D	16	B	36	B	16	A	16	D
17	A	17	D	37	C	17	D	17	A
18	B	18	B	38	D	18	B	18	B
19	E	19	E	39	E	19	E	19	C
20	(*)	20	C	40	E	20	E	20	B

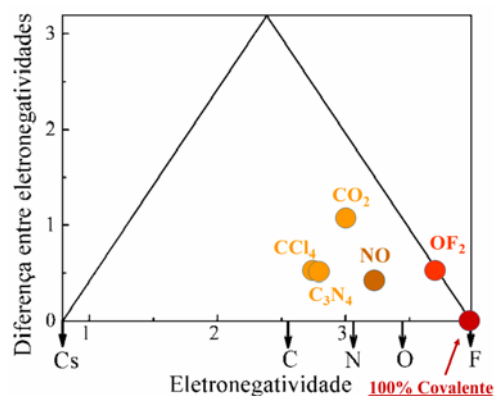
OBSERVAÇÕES

(*) Devido a erros de digitação, as questões 7 e 20 da prova de física foram consideradas corretas para todos os candidatos.

(**) Devido à discordância entre o gabarito oficial do ITA e o apresentado por alguns colégios/cursinhos, segue a resolução oficial do ITA para a questão 09:

RESOLUÇÃO:

Quanto mais próximo um dado composto estiver do vértice direito no diagrama de van Arkel-Ketelaar apresentado, maior será o caráter covalente da ligação química. Assim, não basta analisar apenas o valor da diferença entre eletronegatividades, mas também o valor da eletronegatividade do composto. Os compostos dados no enunciado foram posicionados no diagrama mostrado a seguir, onde se verifica que o composto OF_2 é o que se encontra mais próximo do vértice direito, logo apresenta o maior caráter covalente.



RESPOSTA: OPÇÃO E